

Jade Marie Ducharme

Curriculum Vitae

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Research Interests

I am a radio astronomer specializing in 21-cm experiments and radio-frequency interference mitigation. My research focuses on developing methods to identify, characterize, and mitigate radio-frequency interference in interferometric data, using analytical modeling, Bayesian inference, and machine-learning approaches. I am a member of the MWA Collaboration.

Education

2022–	Brown University <i>Ph.D. (Physics)</i> Advisor: Prof. Jonathan C. Pober
2025	Brown University <i>M.Sc. (Physics)</i>
2022	McGill University <i>B.Sc. (Physics)</i>

Publications

Throughout the following sections, my name is indicated in **bold**.

- (3) Interference Meets Inference: Bayesian Time-Series Modeling of Radio-Frequency Interference
Jade M. Ducharme, Andrei Li, Michael J. Wilensky, and Adrian Liu.
Submitted to ApJ.
- (2) Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies
Jade M. Ducharme, Jonathan C. Pober, and Michael J. Wilensky.
Submitted to PASA.
- (1) Altitude Estimation of Radio Frequency Interference Sources via Interferometric Near-Field Corrections
Jade M. Ducharme & Jonathan C. Pober.
2025, *PASA*, 42, e010. DOI: <https://doi.org/10.1017/pasa.2024.123>.

Collaboration Publications

The following are papers in which my authorship results from participation in a collaboration.

- (3) The Murchison Widefield Array Phase III upgrade: Sensitivity doubled, number of baselines quadrupled, flexibility enhanced, and EoR observations optimised
S.J. Tingay et al., including **J. Ducharme**.
2026, *PASA*, 43, e085. DOI: <https://doi.org/10.1017/pasa.2026.10218>.
- (2) Improved Limits on the 21 cm Signal at $z = 6.5\text{--}7.0$ with the Murchison Widefield Array Using Gaussian Information
Cathryn M. Trott et al., including **J. Ducharme**.
2025, *ApJ*, 991, 211. DOI: <https://doi.org/10.3847/1538-4357/adff80>.
- (1) Limits on the 21 cm Power Spectrum at $z = 6.5\text{--}7.0$ from Murchison Widefield Array Observations
C. D. Nunhokee et al., including **J. Ducharme**.
2025, *ApJ*, 989, 57. DOI: <https://doi.org/10.3847/1538-4357/adda45>.

Conferences and Talks

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| July 2026 | Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies
Talk — MWA Project Meeting 
<i>Brown University, Providence, RI, USA.</i> |
| June 2026 | Improved Modeling for Moving Sources of Radio Frequency Interference and Impact on Flagging Strategies
Poster — Canadian Astronomical Society's Annual General Meeting 
<i>University of Montréal, Montréal, QC, Canada.</i> |
| Feb. 2026 | Mitigating Radio Frequency Interference in Radio Astronomy: Recent Progress at Brown
Invited Talk — MIT Monday Afternoon Science Talks 
<i>MIT Kavli Institute, Cambridge, MA, USA.</i> |
| Jan. 2026 | Altitude estimation of radio frequency interference sources via interferometric near-field corrections
Talk — National Radio Science Meeting 
<i>Boulder, CO, USA.</i> |
| Aug. 2024 | U.S.-Based Research on Mitigating Radio Frequency Interference for MWA Data
Invited Talk — MWA Project Meeting 
<i>Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland.</i> |

Honors and Awards

2026	<i>Brown University</i> Physics Merit Fellowship Full funding for one academic year
2025	<i>Brown University</i> Forrest Award for Excellent Work Related to Experimental Apparatus USD \$250
2024	<i>MWA Collaboration</i> Awarded Best Talk at MWA Project Meeting
2023–2025	<i>National Science Foundation</i> Collaborative Research: 21 cm Reionization Science with the MWA Award #2106510 (full-year academic funding)
2021	<i>McGill Space Institute</i> Summer Research Showcase Prize Winner CAD \$250
2021	<i>Natural Sciences and Engineering Research Council of Canada</i> Undergraduate Student Research Award CAD \$7000

Professional Memberships

2026–	<i>American Astronomical Society (AAS)</i> 
2026–	<i>Canadian Astronomical Society – Société Canadienne d’Astronomie (CASCA)</i> 
2023–	<i>MWA Collaboration</i> 

Teaching and Mentoring

Summer 2026	Co-supervised undergraduate student Neal Frankenberg <i>Brown University, Providence, RI, USA.</i>
Spring 2026	Co-supervised undergraduate student Jacob Lipman <i>Brown University, Providence, RI, USA.</i>
Summer 2025	Co-supervised undergraduate student Andrei Li <i>McGill University, Montréal, QC, Canada.</i>
Spring 2023	Teaching Assistant – PHYS 0220 (Astronomy) <i>Brown University, Providence, RI, USA.</i>
Fall 2022	Teaching Assistant – PHYS 0270 (Introduction to Astronomy) <i>Brown University, Providence, RI, USA.</i>

Professional Development

2026	INTERSECT Research Software Engineering Bootcamp ↗ <i>Princeton University, NJ, USA.</i>
2026	Code/Astro Workshop ↗ <i>UC Santa Cruz, CA, USA.</i>
2024, 2025	AI Winter School ↗ <i>Brown University, Providence, RI, USA.</i>
2024	Precision Calibration Workshop ↗ <i>McGill University, Montreal, QC, Canada.</i>

Technical Skills

- **Programming:** Python, IDL, Stan, Bash, HTML/CSS, JavaScript
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn
- **High-Performance Computing:** SLURM workload management, distributed computing workflows
- **Radio Astronomy Software:** CASA, FHD/eppsiilon, pyuvdata, WSClean
- **Tools:** Git, Linux, React
- **Spoken languages:** French (native), English (fluent)

Public Outreach

2026	Can Radio Astronomy Survive the Modern World? ↗ Invited Talk — Astronomy on Tap. <i>Providence, RI, USA.</i>
2024–	Event coordinator & lead web developer Astronomy on Tap – Rhode Island Chapter (aot-ri.vercel.app). <i>Providence, RI, USA.</i>